DBMS MINI PROJECT · REVIEW 3

R3

TOURISM PACKAGE AND TRAVEL BOOKING MANAGEMENT SYSTEM

Submitted by Jashwanth Sappidi
Roll No. 25WU0102108
Subject Database Management Systems (DBMS)
Academic Year 2026–2027

Project overview & objectives

A database-driven application for the complete tourism booking lifecycle.

ONE SOURCE OF TRUTH

Packages → bookings → payments → refunds

Manages destinations, itineraries, departures, guides, customers, travellers and operational reports.

01 Connect Java / Python to MySQL

02 Implement CRUD and intelligent search

03 Manage bookings, travellers and payments

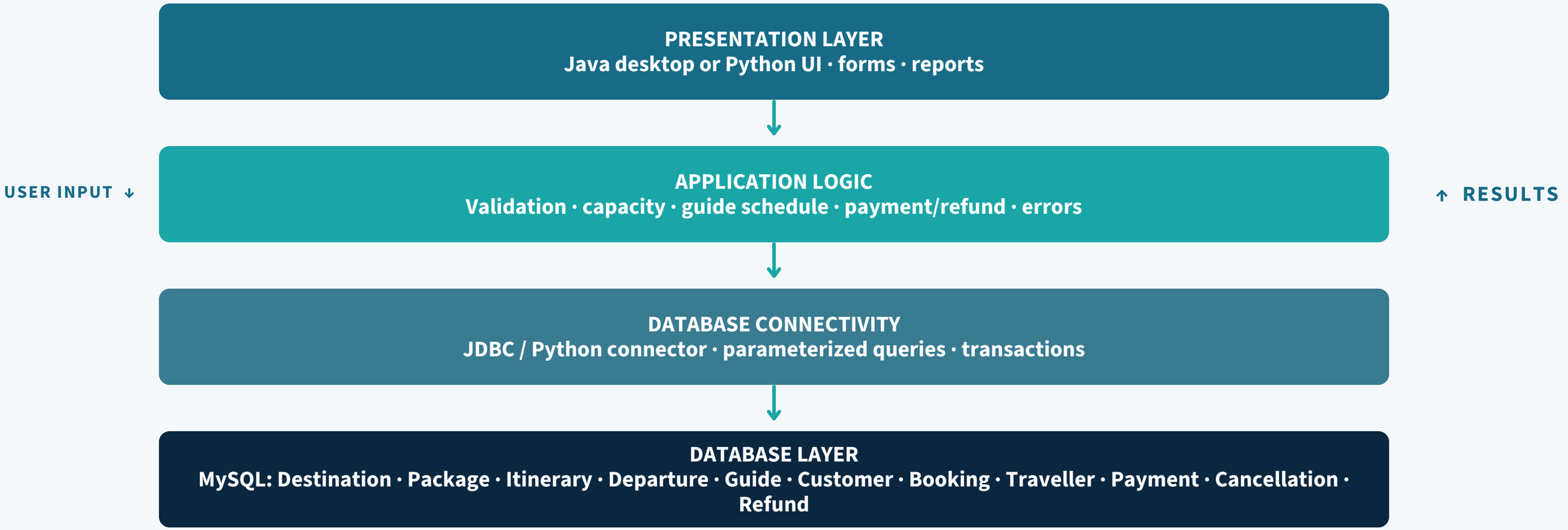
04 Validate capacity and guide schedules

05 Generate decision-ready reports

06 Test integrity and error handling

System architecture

Input flows downward; validated SQL results return upward to the user.

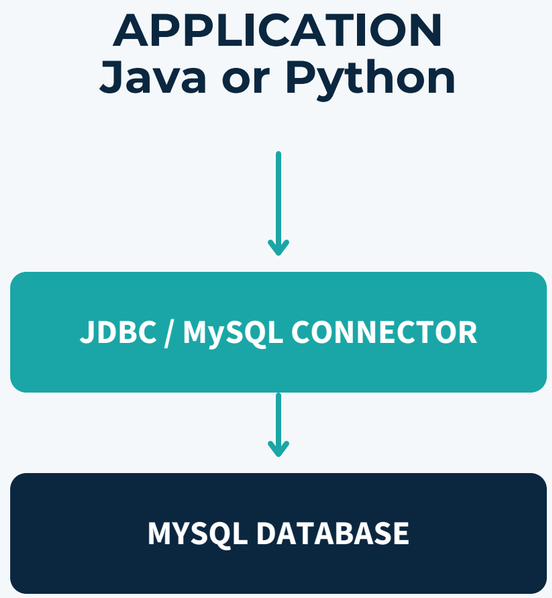


Technologies & database connectivity

Illustrative code — replace credentials and secure them outside source control.

ILLUSTRATIVE · PYTHON

```
import mysql.connector
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="YOUR_PASSWORD",
    database="tourism_db"
)
cursor = conn.cursor()
print("Database connection established")
```



SQL CAPABILITIES

DDL · DML · joins · subqueries · aggregates · views

Package & itinerary management

CRUD packages, search destinations, and enforce unique day numbers per package.

ILLUSTRATIVE PACKAGE FORM

Package name Coastal Discovery

Destination Visakhapatnam

Duration 4 days

Price ₹18,500

ITINERARY

DAY 01 Arrival & city tour

DAY 02 Beach trail & museum

DAY 03 Excursion

DAY 04 Departure

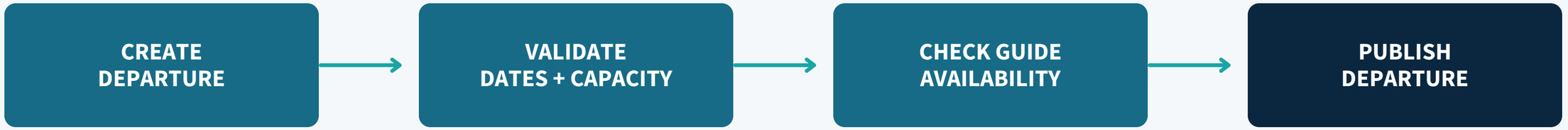
ILLUSTRATIVE SQL

```
SELECT p.package_name, i.day_number, i.activities
FROM Package p
JOIN Itinerary_Day i ON p.package_id = i.package_id
WHERE p.package_id = 101
ORDER BY i.day_number;
```

Sequence is deterministic through ORDER BY day_number.

Departure publishing & guide assignment

A departure moves from draft to confirmed only after every rule passes.



VALIDATION GATE

Capacity > 0

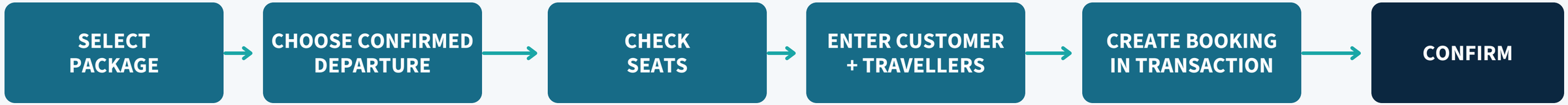
End date ≥ start date

No overlapping guide assignment

Conflict found? Reject assignment, preserve Draft status, and show a clear corrective message.

Customer, traveller & booking management

Transactional booking creation prevents concurrent users from overbooking a departure.



AVAILABLE SEATS

Departure capacity – travellers in confirmed bookings

TRANSACTION GUARD

Lock or safely re-check capacity before commit. If any validation fails, roll back the entire booking.

Payment management

Verified payment means a validated transaction — not merely a successful database insert.



ILLUSTRATIVE SQL · PAYMENT SUMMARY

```
SELECT b.booking_id, c.name, SUM(p.amount) AS total_paid
FROM Booking b JOIN Customer c ON b.customer_id=c.customer_id
JOIN Payment p ON b.booking_id=p.booking_id
WHERE p.status='Paid' GROUP BY b.booking_id, c.name;
```

STATUS

Pending

Paid

Failed

Cancellation & refund management

Transaction-based validation — or carefully designed triggers — protects financial integrity.

REFUND GUARDRAILS

Eligible booking
Non-negative amount
Refund ≤ paid amount
No duplicate refund

ILLUSTRATIVE SQL

```
SELECT c.cancellation_id, c.booking_id,
       r.amount, r.refund_date, r.status
FROM Cancellation c
LEFT JOIN Refund r
  ON c.cancellation_id = r.cancellation_id;
```

Cancel eligible booking → update status → calculate refund → record outcome

CRUD operations & search

Parameterized queries protect inputs while filters make operational data findable.

CREATE
Packages · customers · bookings

READ
History · departures · availability

UPDATE
Permitted records only

DELETE
Constraint-aware removal

PARAMETERIZED SEARCH · PYTHON SYNTAX

```
SELECT package_id, package_name, duration_days, price
FROM Package
WHERE destination_id = %s AND price <= %s;
```

FILTERS + SAFETY

Destination · price · date

Required-field validation

Foreign-key and DB error handling

FINAL DEMONSTRATION

Package + itinerary → departure → guide → customer + travellers → seat check → booking → payment → cancellation / refund → reports

DELIVERABLES

3NF database · DDL scripts · sample data + queries · application source · MySQL connectivity · CRUD + search · validations · test evidence · final report + viva prep

A normalized relational database and front-end application integrate the tourism booking lifecycle while constraints, transactions and validation reduce errors and preserve data integrity.

THANK YOU